

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Multimembrane Inlet System for Mass Spectrometry Analysis

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ABSTRACT

The simulation and experimental tests of the Multimembrane Inlet System that consists of consecutively positioned sheet silicone membranes are described. The original multimembrane inlet supplies preconcentration of volatile organic compounds (VOCs) from inorganic matrices (air, water) during an unstable process of sample injecting into the ion source of a mass spectrometer. The results indicate that silicone membrane of 25-30 micrometers thickness allows to approach an enrichment effect of compounds from matrices by designed inlet system to the maximum of magnitude (for unlimited thin membranes). Our multimembrane inlet system were used for the determination of organic and inorganic compounds in air samples. The effect of preconcentration is $>100\ 000$ by 3-membrane inlet system. The possibilities of using of multimembrane inlet system for environmental application are discussed.